Computers & Operations Research 77 (2017) 210–225

Contents lists available at ScienceDirect

Computers & Operations Research

journal homepage: www.elsevier.com/locate/caor

Performance modeling of a two-echelon supply chain under different levels of upstream inventory information sharing

Sandeep Srivathsan [a,*], Manjunath Kamath [b]

[a] School of Mechanical and Aerospace Engineering, Nanyang Technological University, Singapore
[b] School of Industrial Engineering and Management, Oklahoma State University, Stillwater, OK, USA

ARTICLE INFO

Article history:
Received 18 September 2015
Received in revised form
1 June 2016
Accepted 11 July 2016
Available online 14 July 2016

Keywords:
Queueing models
Supply chain network
Inventory information sharing
Base-stock policy

ABSTRACT

The advancement in information technology has facilitated the sharing of information in supply chain networks (SCNs), resulting in effective management of inventory and storage capacity. In this paper, our focus is on upstream inventory information sharing. Existing analytical performance evaluation models of SCNs are not capable of assessing the impact of inventory information sharing. To address this need, we develop performance evaluation models of SCNs that explicitly consider production capacity, inventory related decisions, variability, transit delays and inventory information sharing in a unified manner. We employ a two-echelon SCN configuration with two retail stores and two production facilities as a test bed. The retail stores have inventory information from the production facilities. We model three levels of inventory information sharing in our study; the information shared ranges from the stock-out information at the lowest level to inventory and backorder level information at the highest level. We develop analytical models first for Poisson arrivals and exponential processing times under all levels of inventory information sharing. We extend these models to general inter-arrival and processing time distributions and subsequently include transit delays between the production facilities and the retail stores. We demonstrate the performance prediction capability of the analytical models developed via extensive numerical experimentation.

© 2016 Elsevier Ltd. All rights reserved.

1. Introduction

The advancement in e-business and e-commerce has meant that customers have more choices for a product as well as retailers have more choices for producers/suppliers of that product. This has forced firms in a supply chain network (SCN) to adopt a more customer-centric view, where the focus is on reducing the stock-out rate and improving customer satisfaction. This need was clearly highlighted by Corsten and Gruen [4] in their survey-based study. The study found that the average stock-out rate over 40 stores was 8.3%, and that customers reacted to this situation by switching the store or brand, delaying the purchase at the same store, or by not purchasing the item.

The advancement in information technology has facilitated enhanced information sharing among firms in a SCN, resulting in better management of their inventory and storage capacity. In this paper, we consider upstream inventory information sharing as a means to alleviate the stock-out problem. Performance evaluation tools for SCNs can aid in understanding the complex dynamics of the system, and in explaining how the SCN performance measures (e.g. fill rate, expected inventory level, and expected number of backorders) respond to changes in the SCN parameter values (e.g. demand inter-arrival time and processing time parameters and base-stock levels). However, the existing performance evaluation models of SCNs are not capable of studying the impact of upstream inventory information sharing. To address this need, we focus on the development of analytical performance evaluation tools to assess the impact of sharing upstream inventory information on SCN performance. These tools can be used in tandem with optimization models to determine key business parameters such as target inventory levels of raw material, intermediate goods, and finished goods as well as capacities of production facilities.

In our study, we employ a "building-block" type SCN configuration composed of two retail stores and two identical production facilities (same base-stock level and processing time distributions). The inter-arrival times between demands at the retail stores and processing times at the production facilities, as well as the transit time between the production facilities and retail stores, follow a general distribution. Inventory information shared by the production facilities with the retail stores can range from stock-out information at the lowest level to the inventory and backorder

* Corresponding author.
E-mail addresses: ssandeep@ntu.edu.sg (S. Srivathsan), m.kamath@okstate.edu (M. Kamath).

http://dx.doi.org/10.1016/j.cor.2016.07.011
0305-0548/© 2016 Elsevier Ltd. All rights reserved.

information at the highest level. The retail store is assumed to fully exploit the information available in their order routing strategy. For example, when the inventory and backorder levels at the production facility are available, the retail store always places an order with the production facility with the least number of outstanding orders.

We first used the Markov chain approach to analyze the SCN under Poisson arrivals and exponential processing times. We solved the Markov chain numerically by limiting the maximum number of backorders. We then developed approximate queueing models for the SCN under Poisson arrivals and exponential processing times using the following insights. A careful examination of the order routing policy for the highest level of information sharing from the perspective of the production facility revealed similarities with the experience of a server in a multi-server queue (see Section 4.1 for more details). This led to the development of the analytical model for SCN with the highest level of information sharing. In the case of lower levels of information sharing, we used insights from the Markov chain model and simulation results to obtain an analytical model. These models were extended to general inter-arrival and processing time distributions, and then to include transit delays between the production facilities and the retail stores. We demonstrate the performance prediction accuracy of the analytical models through extensive numerical experimentation, wherein the analytical results are compared to results from simulation models.

The remainder of the paper is organized as follows. Section 2 first reviews the literature on modeling production-inventory networks and SCNs and then summarizes research studies on information sharing in SCNs. Section 3 describes the structure of the SCN used in our study. Section 4 presents the analytical models developed for the SCN under Poisson arrivals and exponential processing times for all levels of inventory information sharing, while the analytical models for the SCN under general inter-arrival and processing time distributions are presented in Section 5. Extensions to incorporate transit delays are presented in Section 6. Finally, the conclusions are presented in Section 7.

2. Literature review

The early literature on queueing models of production-inventory networks (PINs) includes the work of Buzacott and Shanthikumar [3] who modeled various single-stage production-inventory systems, and the efforts of Lee and Zipkin [14] who focused on multi-stage tandem PINs. A detailed review and comparison of performance evaluation models of PINs is presented in Kamath and Srivathsan [9].

The early literature on performance evaluation of SCNs include Lee and Billington [13] and Ettl et al. [5]. While Lee and Billington [13] focused on performance evaluation to manage the material flows in decentralized supply chains, Ettl et al. [5] focused on strategic level decision making in large SCNs using optimization and queueing based models in tandem. Srivathsan [17] extended the PIN models developed by Sivaramakrishnan [16] to analyze a generic supply chain network by modeling supply, transportation and distribution operations. The backorder distribution was used to link successive stages of the SCN within the solution approach [17,19].

Li et al. [15] presented a detailed review of studies on the value of information sharing in supply chains. Some of the research studies that focus on inventory information sharing include Armony and Plambeck [2], Gavirneni [6], and Zipkin [25]. Gavirneni [6] used infinitesimal perturbation analysis to study a periodic review inventory control problem in a supply chain network. The network consisted of one capacitated supplier who supplies a single product to multiple identical retailers. The study considered three levels of cooperation – no cooperation, medium cooperation (sharing of retailers' current inventory level) and complete cooperation (additional information about possibility of transferring inventory between retailers). The computational results showed that the third level of coordination resulted in the least cost and the benefits of cooperation in this supply chain decreased with increase in supplier capacity, increase in number of retailers, decrease in penalty cost, and decrease in demand variance. Armony and Plambeck [2] studied the effect of duplicate orders on the manufacturer's estimation of demand rate and customer's sensitivity to delay for a SCN with one manufacturer and two distributors. Zipkin [25] studied the performance of a multi-item production-inventory system with a production facility under the first-come-first-served (FCFS) policy and longest-queue (LQ) policy (orders were satisfied based on the longest backorder queue). The study found that the LQ policy outperformed the FIFO policy by about 20% in some cases. The difference in performance was the greatest when the number of product types was large and variability in inter-arrival and processing time distributions was low. The difference was small at low utilization levels, increased until 90% utilization and vanished in heavy traffic.

Many of the studies that deal with inventory information sharing [2,6,25] consider downstream inventory information sharing (retail store sharing information with a production facility). Our work focuses on modeling upstream inventory information sharing (production facility sharing information with the retail store). The theoretical contribution of our work is in the development of performance evaluation models that explicitly consider production capacity, inventory related decisions (e.g. base-stock levels), variability (in processing times and demand arrivals), transit delays (inventory in transit) and information sharing in a unified manner. The existing studies on production-inventory systems mostly deal with incorporating capacity and inventory issues within a single model. In contrast, our model can also explicitly model the sharing of inventory information. With the ability to consider inventory information sharing at different levels, our models can be used to evaluate how the performance measures such as fill rate and expected number of backorders are affected by any change in the SCN parameters such as demand inter-arrival time and processing time parameters, base-stock levels and level of information sharing.

3. SCN structure

We consider a two-echelon SCN with two retail stores and two symmetrical production facilities (same base-stock level and processing time distribution), each with its own output store to stock finished products. Each store in the SCN is assumed to operate under a base-stock control policy with one-for-one replenishment. The base-stock level is R_i at retail store i $(i = 1, 2)$ and S at each production facility. Each customer order is assumed to be for a single unit of the finished product. The demand inter-arrival times at the retail stores as well as the processing times at the production facilities follow general distributions. The arrival of a demand consumes a finished product at the retail store if it is available and causes an order for replenishment to be placed at one of the two production facilities (see Fig. 1). If available, a finished product from the output store of the production facility is immediately shipped to the retail store and the output store sends an order for replenishment to its processing stage. In this paper, we assume that the processing stage has a single server. An order can join the Work in Process (WIP) queue or be processed as soon as it is received. There are no limits on the number of backorders at either production facility.

We consider three levels of upstream inventory information sharing that vary in the amount of information shared by a production facility with the retail stores. In practice, this level would depend on factors such as trust and supply contracts. The order

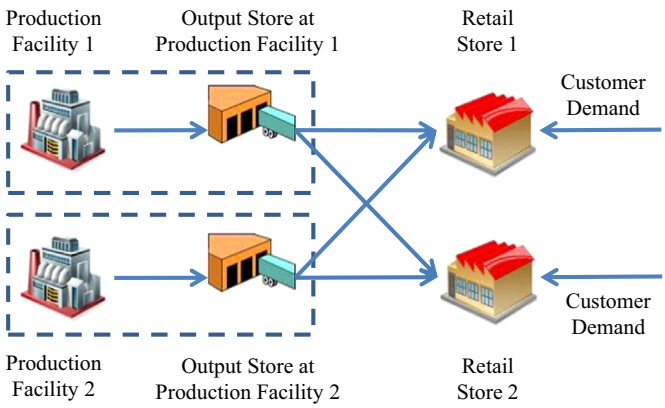

Fig. 1. SCN structure.

routing mechanism in SCNs with upstream inventory information sharing can be significantly different when compared with SCNs with no inventory sharing. In the latter case, the orders from a retail store could be routed to the production facilities according to fixed probabilities (based on preference or historical information). This can sometimes result in an order being placed at a production facility facing a stock-out situation even when there is inventory at the other facility. In a SCN with different levels of information sharing, this situation can be completely avoided as shown in Table 1

In the case of the SCN with a high level of information sharing (HiIS), we note from Table 1 that the orders are always routed based on the highest number of items in stock when there is inventory at both facilities and the shortest backorder queue when both facilities are backordered. Also, when the base-stock levels at the two production facilities are the same, the HiIS routing policy closely resembles the shortest queue problem that has been studied in the literature (see e.g. [1,8,11]).

The following notations are used in this paper. Since the two production facilities are identical, for simplicity we drop the subscripts representing the identity of a production facility.

Model parameters

λ_i^r	Order arrival rate at retail store i
ca_i^2, cs^2	Inter-arrival time SCV[a] at retail store i and processing time SCV[a] at a production facility, respectively
R_i, S	Base-stock level at retail store i and output store of a production facility, respectively

[a] SCV or squared coefficient of variation = variance/(mean)^2.

Model variables (uppercase letters are random variables)

N_i^r	Number of orders at retail store i
N^p	Number of orders at a production facility
ρ	Utilization of a production facility
I_i^r, I^p	Inventory level at retail store i and output store of a production facility, respectively
B_i^r, B^p	Number of backorders at retail store i and at a production facility, respectively
f_i^r, f^p	fill rate at retail store i and at the output store of production facility, respectively
W_i^{br}, W^{bp}	time to fulfill a backorder at retail store i and at a production facility, respectively
T_i^r, T^p	time spent by an order at retail store i and at a production facility including its output store, respectively

4. Analytical models for Poisson arrivals and exponential processing times under different levels of inventory information sharing

First, we explored the development of a continuous time Markov chain (CTMC) model in which the state is given by $X(t) = \{i, j, k, l\}$, where i, j, k and l are non-negative integers representing the number of orders (replenishment plus backorders) at retail stores 1 and 2, and production facilities 1 and 2, respectively. Since there is no limit on the number of backorders, the state space of the CTMC is infinite. Furthermore, the order placement policies resulted in numerous conditions for defining the state transitions and the balance equations. Only an approximate solution was possible by numerically solving the CTMC with arbitrary limits on backorders (for additional details see [18]). As a result, we focused on developing an approximate queueing-based model for the SCN with HiIS, and then modified this model for SCN with lower levels of inventory information sharing using a combination of analytical reasoning and empirical observations. Section 4.1 presents the analytical model developed for the SCN with HiIS, while Section 4.2 presents its modification for lower levels of information sharing (MedIS and LoIS).

4.1. Analytical model for SCN with HiIS

As discussed in the previous section, the routing strategy used under HiIS is equivalent to choosing the shortest queue when the two production facilities have identical base-stock levels. No closed form solution exists for the shortest queue problem [1]. Furthermore, in the model for the SCN we need information about the retail store that placed the order to obtain the distribution of the number of outstanding orders at the individual retail stores. We developed

Table 1
Levels of information sharing – information shared and order routing strategy.

Level of inventory information sharing	Inventory information shared	Order routing strategy based on status of production facility		
		Both facilities have inventory	Both facilities have backorders	Only one facility has inventory
Low (LoIS)	1. Presence or absence of inventory	Equal probability	Equal probability	Facility with inventory
Medium (MedIS)	1. Presence or absence of inventory 2. Backorder level	Equal probability	Facility with least backorder	Facility with inventory
High (HiIS)	1. Inventory level 2. Backorder level	Facility with maximum inventory	Facility with least backorder	Facility with inventory

an approximate analytical model based on the following observations. In the SCN with HiIS, the retail store orders are not routed to a production facility with an empty output store, when the other production facility has items in stock. In addition, the orders are always routed to the production facility with the smaller number of orders. Note that the production facility with the smaller number of orders will have the maximum inventory level when there is inventory at both stores and the lower backorder level when both stores are backordered. This routing policy results in balancing the number of orders at the two production facilities. From the perspective of the processing stage at a production facility, this closely resembles the experience of a server in a multi-server queue (M/M/2 queue in the case of two production facilities) where an arriving customer (order in our case) waits in a common queue before joining the first available server. Additionally, a server in a multi-server queue can never be idle when there are customers waiting to be processed. This means that some of the states in the exact CTMC model for HiIS case, namely, $(i, j, k, 0)$ and $(i, j, 0, l) \ni k + l \geq 2$ are not modeled in an M/M/2 system. But the order routing in the HiIS case ensures that the steady state probabilities of the system being in these states are extremely small. With this knowledge, we used the M/M/2 queue to develop an approximation for the SCN with HiIS [20]. The advantage of using this approximation is that closed form expressions can be obtained for the performance measures at all stores and the model can be easily extended to consider SCN with more than two production facilities.

Fig. 2 shows the exact representation of our SCN, while Fig. 3 shows the aggregate representation based on the approximation. In essence, the two production facilities are modeled as an M/M/2 queueing system, where the net arrival process is the superposition of the two arrival streams from the retail stores (i.e. $\lambda = \lambda_1^r + \lambda_2^r$ and the combined arrival process is also Poisson). Fig. 3 shows that in approximate model, the inventory at the two production facilities would be "pooled" and this leads to some loss of information about inventory and orders at the individual production facilities.

To clearly illustrate the difference between the original and the pooled models, we map the net inventory position (NIP) at the two production facilities to the states of the servers in the two models as shown in Table 2. In the following table, let α_1 represent the probability that an order is placed at production facility 1. Also note that NIP would be positive when the facility has inventory and negative when the facility is backordered.

Since the distribution of the number of orders and the performance measures at the individual production facilities are obtained from the corresponding values at the pooled facility, we use the following notations for the pooled facility:

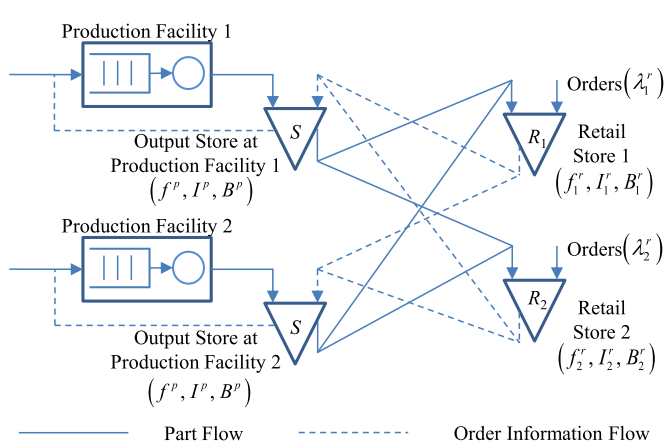

Fig. 2. Exact representation of the SCN structure.

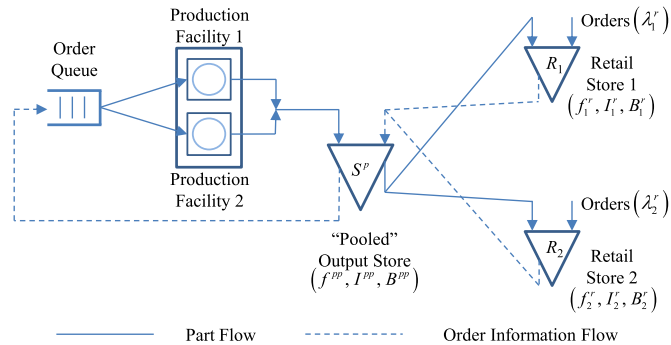

Fig. 3. Aggregate representation of the SCN structure.

Table 2
Mapping of net inventory position to state of the servers.

NIP (i, j)	State of servers (original model)	α_1	States of the servers (pooled model)
$i = j = S$	Servers at both facilities are idle	0.5	Both servers are idle
$i = j < S$	Servers at both facilities are busy	0.5	Both servers are busy
$i < j = S$	Server at facility 2 is idle	0	One server busy if $i = S - 1$, both servers are busy otherwise
$i < j < S$	Servers at both facilities are busy	0	Both servers are busy
$S = i > j$	Server at facility 1 is idle	1	One server busy if $j = S - 1$, both servers are busy otherwise
$S > i > j$	Servers at both facilities are busy	1	Both servers are busy

Model parameters	
λ	Order arrival rate at the pooled facility
cap^2	Inter-arrival time SCV at the pooled facility
S^p	Base-stock level at the output store of the pooled facility; $S^p = 2S$.

Model variables (uppercase letters are random variables)	
N^{pp}	Number of orders at the pooled facility
Q^{pp}	Number of orders in queue at the pooled facility
I^{pp}	Inventory level at the output store of the pooled facility
B^{pp}	Number of backorders at the output store of the pooled facility
f^{pp}	fill rate at the output store of the pooled facility

When an order is placed at a retail store, an instantaneous replenishment order is placed at the pooled output store. Since transit time is not modeled (this assumption will be relaxed later; see Section 6), if a finished part is available at the output store of the pooled facility, the part will be instantaneously available at the retail store. As a result, an order (replenishment or backorder) at a retail store can be reflected only as a backorder at the pooled facility. It has to be noted that an order at the pooled facility is an "aggregate" order; the identity of the retail store that placed the order is lost while merging the two arrival streams from the retail stores. Hence, a "disaggregation" procedure was used to find the distribution of the number of orders pertaining to retail store i, N_i^r, at the pooled facility by conditioning on the number of backorders, B^{pp}, at the pooled facility. This conditional distribution is Binomial as shown in Eq. (1). The parameters of the binomial distributions

are the proportions of orders from retail store i, γ_i, and l, the total number of backorders at the pooled facility. Now the unconditional probability that a backorder at the pooled facility belongs to retail store i can be easily obtained by multiplying the above conditional probability by the probability of backorders and summing over all values of l (backorders) greater than or equal to m (number of orders at a retail store) as shown in Eq. (3) [18]. The backorder distribution at the pooled facility can be obtained from the distribution of the number of orders at pooled production facility N^{pp} as shown in Eq. (2). The distribution of N^{pp} is shown later in Eq. (10):

$$P(N_i^r = m|B^{pp} = l) = \binom{l}{m}\gamma_i^m(1-\gamma_i)^{(l-m)}; \quad m = 0, 1, \ldots, l, \quad i = 1, 2, \tag{1}$$

where $\gamma_i = \lambda_i^r/(\lambda_1^r + \lambda_2^r)$; $i = 1, 2$.

$$P(B^{pp} = l) = \begin{cases} \sum_{k=0}^{S^p} P(N^{pp} = k); & l = 0 \\ (N^{pp} = S^p + l); & l \geq 1 \end{cases}. \tag{2}$$

$$P(N_i^r = m) = \sum_{l \geq m} P(N_i^r = m|B^{pp} = l) \cdot P(B^{pp} = l) = \begin{cases} \left[1 + 2\rho\left(\frac{1-\rho^{2S}}{1-\rho}\right)\right] + \left[\frac{2\gamma_j\rho^{2S+1}}{1-\gamma_j\rho}\right]\left(\frac{1-\rho}{1+\rho}\right); & m = 0 \\ \left[\frac{2\gamma_i^m\rho^{2S+m}}{\left[1-\gamma_j\rho\right]^{m+1}}\right]\left(\frac{1-\rho}{1+\rho}\right); & m \geq 1 \end{cases}, \tag{3}$$

In Eq. (3), j takes a value of 1 when the retail store index i is 2, and a value of 2 when the retail store index i is 1. Note that Eq. (3) is applicable only for a SCN with two identical production facilities. The retail store performance measures that are of interest include the fill rate f_i^r, expected number of backorders $E[B_i^r]$, expected inventory level $E[I_i^r]$, expected time to fulfill a backorder $E[W_i^{br}]$, and the expected time spent by an order $E[T_i^r]$. We used the ready rate (probability that an item is available) to approximate the fill rate at all stages. Note that this approximation would be exact only when the arrival process is Poisson (PASTA principle [24]). The performance measures at retail store i, i=1, 2, are presented in the following equations:

$$f_i^r = \sum_{j=0}^{R_i-1} P(N_i^r = j), \tag{4}$$

$$E[I_i^r] = \sum_{j=0}^{R_i-1} (R_i - j)P(N_i^r = j), \tag{5}$$

$$E[N_i^r] = \sum_{j \geq 1} jP(N_i^r = j), \tag{6}$$

$$E[B_i^r] = E[I_i^r] + E[N_i^r] - R_i, \tag{7}$$

$$E[W_i^{br}] = \frac{E[B_i^r]}{(1 - f_i^r)\lambda_i^r}, \tag{8}$$

$$E[T_i^r] = \frac{R_i + E[B_i^r]}{\lambda_i^r}. \tag{9}$$

Since the pooled facility is modeled as an M/M/2 queueing system, the distribution of the number of orders at the pooled facility, N^{pp}, is given by Eq. (10) (see [7] for more details). The two production facilities are identical and we exploit this symmetry in the calculation of the fill rate f^p, expected backorders $E[B^p]$, the expected inventory $E[I^p]$, and the expected time to fulfill a backorder $E[W^{bp}]$ at the individual production facilities. First, an item in the output store of the pooled facility in the aggregate representation is equally likely to be in the output store of either production facility. Hence, the expected number of backorders and the expected inventory level at a production facility is half of the corresponding averages for the pooled facility as shown in Eqs. (12) and (13). Regarding fill rates, we can relate the fill rate at the pooled facility to the fill rates at the individual production facilities as follows:

$$f^{pp} = P(\textit{order goes to production facility } 1).\ \textit{fill rate of production facility } 1 + P(\textit{order goes to production facility } 2).\ \textit{fill rate of production facility } 2$$

Now because of symmetry the two probabilities are the same and equal to 0.5, and the fill rates at the production facilities are the same. Hence we have

$$f^{pp} = 0.5.\ f^p + 0.5.\ f^p = f^p.$$

All performance measures at a production facility can be obtained from Eqs. (11)–(15):

$$P(N^{pp} = k) = \begin{cases} \frac{(2\rho)^k}{k!}P(N^{pp} = 0); & k = 1 \\ 2\rho^k P(N^{pp} = 0); & k \geq 2 \end{cases}, \tag{10}$$

where $\rho = \frac{\lambda}{2\mu}$, $P(N^{pp} = 0) = \frac{1-\rho}{1+\rho}$.

$$f^p = \sum_{k=0}^{S^p-1} P(N^{pp} = k) = \frac{1 + \rho - 2\rho^{2S}}{1 + \rho}, \tag{11}$$

$$E[I^p] = 0.5\sum_{k=0}^{S^p-1}(S^p - k)P(N^{pp} = k) = \frac{S(1-\rho)^2 + \rho(2S + \rho^{2S} - 2S\rho - 1)}{1 - \rho^2}, \tag{12}$$

$$E[B^p] = 0.5\sum_{k \geq S^p}(k - S^p)P(N^{pp} = k) = \frac{\rho^{2S+1}}{1 - \rho^2}, \tag{13}$$

$$E[W^{bp}] = \frac{E[B^p]}{(\lambda/2)(1 - f^p)} = \frac{\rho}{\lambda(1 - \rho)}, \tag{14}$$

$$E[T^p] = \frac{S + E[B^p]}{\lambda/2}. \tag{15}$$

4.2. Analytical model for SCN with lower levels of inventory information sharing (LoIS and MedIS)

In this section, we develop a modified analytical model to better predict the performance measures at the two lower levels of inventory information sharing (LoIS and MedIS). This modification involves the development of a correction factor for the steady state probabilities. To get a better idea about the steady-state probability distributions, we considered a CTMC model with a state-space defined by (i, j), where i and j represent the number of outstanding orders at the two production facilities. Since the production facilities are symmetrical, we obtained a reduced Markov chain by combining states (i, j) and (j, i). The reduced CTMC model revealed some state transition patterns, but we could not exploit these patterns to arrive at a closed-form solution.

Hence, we fixed the maximum number of backorders at the retail store to 1000 and numerically solved the CTMC (see [18] for additional details). The modification to the steady state probabilities is based on a combination of analytical reasoning and empirical observations as explained below:

1. As the base-stock level is increased, the routing policies for LoIS and MedIS behave more like the (Bernoulli) routing policy for the SCN with no information sharing. This is because of the fact that there is always enough inventory in stock to satisfy the customer orders and the stock-out probability becomes negligibly small.
2. In the analytical model for HiIS, $\pi_{0,0}$ or π_0 (the steady-state probability that there are no orders at both production facilities) depends only on the utilization and is independent of the base-stock level S. However, simulation and CTMC results indicated that the $\pi_{0,0}$ value is dependent on both the base-stock level, S, and the utilization, ρ, for the LoIS and MedIS cases. The reason for this dependence is the probabilistic routing of orders in the presence of inventory at both production facilities. Note that the structure of the Markov chain differs for each value of S in these cases [18]. In the HiIS case, $\pi_{0,0}$ is dependent only on ρ because of the fact that irrespective of the base-stock level, the orders are always routed to the store with the maximum inventory when there is inventory at both production facilities and to the store with minimum number of backorders in the presence of backorders at both production facilities. The equivalent probability value in the HiIS case is only dependent on the utilization ($P(N^{pp}=0)=(1-\rho)/(1+\rho)$).
3. As the base-stock level is increased, the value of $\pi_{0,0}$ in the LoIS and MedIS cases approaches $(1-\rho)^2$ (corresponding value for the Bernoulli routing). Further, the value of $\pi_{0,0}$ is always greater than $(1-\rho)^2$ as seen in Table 3.

The above observations formed the basis for the development of a correction factor for the probability $P(N^{pp}=0)$. Table 3 illustrates the dependency of $\pi_{0,0}$ for LoIS and MedIS on the base-stock level and the utilization. Further, $\pi_{0,0}$ becomes nearly identical for the LoIS and MedIS cases as the base-stock level increases. Since the value of $\pi_{0,0}$ is always greater than $(1-\rho)^2$, a correction factor of the form $(1-\rho)^2(1+\omega^M(\rho,S))$ was used. It is clear that the $\omega^M(\rho,S)$ term should vanish as the base-stock level increases and approaches infinity. Examples of such a term are an exponential decay function and a multiple of the reciprocal of the base-stock level. To obtain a better idea, we solved the CTMC for LoIS to obtain the $\pi_{0,0}$ for base-stock levels of 2–20 in steps of 2 at the three utilization levels of 70%, 80% and 90% and found that the $\omega^M(\rho,S)$ term can be approximated by an exponential decay function that depends on the utilization of the production facility and its base-stock level. With this information, we found the exponential decay function empirically and the function is presented in Eq. (16) and the scaled value of $P(N^{pp}=0)$ is presented in Eq. (17):

$$\omega^M(\rho,S)=\frac{\exp^{(\rho^2-\sqrt{\rho})S}}{2(1-\rho^2)}, \tag{16}$$

Table 3
Dependence of $\pi_{0,0}$ on the base-stock level and production facility utilization.

ρ	Model	$S=2$	$S=4$	$S=6$	$S=8$	$S=10$	$S=\infty$
70%	MedIS	0.139	0.113	0.101	0.096	0.093	0.090
	LoIS	0.135	0.113	0.101	0.096	0.093	0.090
80%	MedIS	0.082	0.063	0.054	0.048	0.045	0.040
	LoIS	0.078	0.063	0.054	0.048	0.045	0.040
90%	MedIS	0.037	0.027	0.021	0.018	0.016	0.010
	LoIS	0.032	0.026	0.021	0.018	0.016	0.010

Table 4
Experiments for the SCN.

Parameters	Levels	Parameter values
Demand arrival rates $(\lambda_1^r, \lambda_2^r)$	1	(2, 1)
Base-stock levels (S, R_1, R_2)	4	(2, 2, 4)[a], (2, 4, 2)[a], (4, 4, 6), (4, 6, 4)
Utilization level ρ	2	80%, 90%
Inter-arrival time distribution at the retail stores	3	Erlang ($ca_i^2=0.25$), Exponential ($ca_i^2=1.00$)[a], 2-phase hyper-exponential ($ca_i^2=4.00$)
Processing time distribution at the production facilities	3	Erlang ($cs^2=0.25$), Exponential ($cs^2=1.00$)[a], 2-phase hyper-exponential ($cs^2=4.00$)

[a] Parameter values used only for Poisson arrivals and exponential processing times.

$$P(N^{pp}=0)=(1-\rho)\left[\frac{2\left(1-\rho^2\right)+\exp^{(\rho^2-\sqrt{\rho})S}}{2(1+\rho)}\right]. \tag{17}$$

Since the $P(N^{pp}=0)$ value has been modified, there is a need to re-normalize the remaining probabilities in the M/M/2 distribution. The easiest way to achieve this is to scale all the remaining probabilities by a normalization factor $\zeta^M(\rho,S)$. However, further examination of the CTMC models for LoIS and MedIS cases revealed that the probability of having one order in the system is equal to the utilization times $\pi_{0,0}$ (i.e. $\pi_{1,0}=\pi_{0,1}=\rho\pi_{0,0}$). As a result, the probability $\pi_1=(\pi_{1,0}+\pi_{0,1})$ is fixed to the value $2\rho P(N^{pp}=0)$. The terms in the total probability equation are presented in Eq. (18) and the resulting normalization factor $\zeta^M(\rho,S)$ is presented in Eq. (19). The resulting distribution for the random variable representing the number of orders at the pooled production facility, N^{pp}, is presented in Eq. (20):

$$P(N^{pp}=0)\left[1+2\rho+2\rho^2\zeta^M(\rho,S)+2\rho^3\zeta^M(\rho,S)+\cdots\right]=1\ , \tag{18}$$

$$\zeta^M(\rho,S)=\frac{(1+\rho)}{\rho\left[2\left(1-\rho^2\right)+\exp^{-(1-\rho^2)S}\right]}-\frac{(1+2\rho)(1-\rho)}{2\rho^2}, \tag{19}$$

$$P(N^{pp}=k)=\begin{cases}(2\rho)P(N^{pp}=0) & ;\ k=1\\ 2\rho^k\zeta^M(\rho,S)P(N^{pp}=0) & ;\ k\geq 2\end{cases}. \tag{20}$$

The performance measures at a production facility were obtained using Eqs. (21)–(25). The distribution of the number of orders at a retail store was obtained using Eq. (26), and Eqs. (27)–(31) were used to obtain the performance measures at a retail store:

$$f^p=\left[(1+2\rho)(1-\rho)+2\rho^2\zeta^M(\rho,S)\left(1-\rho^{2S-2}\right)\right]P(N^{pp}=0), \tag{21}$$

$$E[I^p]=0.5\sum_{k=0}^{S^p-1}(S^p-k)P(N^{pp}=k), \tag{22}$$

$$E[B^p]=\frac{\rho^{2S+1}\zeta^M(\rho,S)}{(1-\rho)^2}P(N^{pp}=0), \tag{23}$$

$$E[W^{bp}]=\frac{\rho}{(1-\rho)\lambda}, \tag{24}$$

$$E[T^p]=\frac{\left(S+\frac{\rho^{2S+1}\zeta^M(\rho,S)}{(1-\rho)^2}P(N^{pp}=0)\right)}{\lambda/2}. \tag{25}$$

Table 5
Fill rate and expected number of backorders at retail store 1.

(S, R_1, R_2)	ρ	f_1^r					$E[B_1^r]$				
		SimL	SimM	AnL&M	SimH	AnH	SimL	SimM	AnL&M	SimH	AnH
(2, 2, 4)	0.8	0.730	0.727	0.727	0.739	0.759	0.846	0.743	0.729	0.704	0.642
	0.9	0.456	0.454	0.455	0.465	0.493	3.853	3.375	3.271	3.273	3.044
(2, 4, 2)	0.8	0.848	0.854	0.855	0.861	0.873	0.450	0.395	0.386	0.372	0.340
	0.9	0.579	0.596	0.600	0.604	0.627	2.875	2.500	2.403	2.416	2.237
(4, 4, 6)	0.8	0.935	0.932	0.937	0.943	0.948	0.190	0.181	0.167	0.152	0.139
	0.9	0.730	0.721	0.732	0.743	0.755	1.757	1.691	1.607	1.567	1.468
(4, 6, 4)	0.8	0.965	0.964	0.967	0.970	0.972	0.100	0.096	0.088	0.081	0.074
	0.9	0.798	0.795	0.803	0.810	0.820	1.293	1.246	1.181	1.157	1.078

Table 6
Expected inventory level and expected time to fulfill a backorder at retail store 1.

(S, R_1, R_2)	ρ	$E[I_1^r]$					$E[W_1^{br}]$				
		SimL	SimM	AnL&M	SimH	AnH	SimL	SimM	AnL&M	SimH	AnH
(2, 2, 4)	0.8	1.244	1.314	1.351	1.344	1.428	1.567	1.358	1.333	1.347	1.333
	0.9	0.689	0.789	0.819	0.813	0.901	3.539	3.091	3.000	3.058	3.000
(2, 4, 2)	0.8	2.848	2.967	3.001	3.013	3.126	1.479	1.349	1.333	1.334	1.333
	0.9	1.711	1.913	1.951	1.955	2.093	3.407	3.087	3.000	3.048	3.000
(4, 4, 6)	0.8	3.485	3.520	3.571	3.598	3.642	1.455	1.339	1.333	1.343	1.333
	0.9	2.484	2.551	2.630	2.667	2.749	3.257	3.029	3.000	3.047	3.000
(4, 6, 4)	0.8	5.397	5.435	5.492	5.527	5.577	1.416	1.334	1.333	1.335	1.333
	0.9	4.020	4.107	4.203	4.257	4.359	3.198	3.030	3.000	3.047	3.000

Table 7
Expected time spent by an order at retail store 1.

(S, R_1, R_2)	ρ	$E[T_1^r]$				
		SimL	SimM	AnL&M	SimH	AnH
(2, 2, 4)	0.8	1.422	1.371	1.364	1.351	1.321
	0.9	2.924	2.686	2.635	2.635	2.522
(2, 4, 2)	0.8	2.224	2.196	2.193	2.185	2.170
	0.9	3.435	3.248	3.201	3.206	3.118
(4, 4, 6)	0.8	2.097	2.092	2.083	2.077	2.070
	0.9	2.881	2.847	2.804	2.785	2.734
(4, 6, 4)	0.8	3.052	3.050	3.044	3.042	3.037
	0.9	3.649	3.626	3.590	3.580	3.539

$$P(N_i^r = j) = \begin{cases} \left[1 + 2\rho + \dfrac{2\rho^2\zeta^M(\rho, S)\left(1 - \rho^{2S-1}\right)}{(1-\rho)}\right] P(N^{pp} = 0); & j = 0 \\ 2\rho^{2S+j}\zeta^M(\rho, S)P(N^{pp} = 0); & j \geq 1 \end{cases}. \tag{26}$$

$$f_i^r = \left[P(N_i^r = 0) + \frac{2\rho^{2S+1}\zeta^M(\rho, S)\left(1 - \rho^{R_i-1}\right)}{(1-\rho)} P(N^{pp} = 0)\right], \tag{27}$$

$$E[I_i^r] = R_i P(N_i^r = 0) + 2\rho^{2S+1}\zeta^M(\rho, S)P(N^{pp} = 0)\left[\frac{R_i - R_i\rho + \rho^{R_i} - 1}{(1-\rho)^2}\right], \tag{28}$$

$$E[B_i^r] = \frac{2\rho^{R_i+1}\zeta^M(\rho, S)}{(1-\rho)^2} P(N^{pp} = 0), \tag{29}$$

$$E[W_i^{br}] = \frac{\rho}{(1-\rho)\lambda}, \tag{30}$$

$$E[T_i^r] = \frac{\left[R_i + \dfrac{2\rho^{R_i+1}\zeta^M(\rho, S)}{(1-\rho)^2} P(N^{pp} = 0)\right]}{\lambda}. \tag{31}$$

4.2.1. Validation of the analytical models

In this section, the analytical models for the HiIS case and the lower levels of inventory information sharing are validated by comparing the performance measures obtained from the analytical models with simulation estimates. The performance measures were estimated using a simulation model developed in Arena [10]. The warm-up period of 10,000 time units of the simulation model was determined using Welch's method [21] and the simulation model was run for 10 independent replications with the length of each replication set to 100,000 time units. We have considered a robust design of experiments by considering a wide range of values for the SCN parameter values. In Table 4, we present the SCN parameter values that were used for numerical experimentation in Sections 4 and 5. The design consists of

Table 8
Fill rate and expected number of backorders at a production facility.

(S, R_1, R_2)	ρ	f^p					$E[B^p]$				
		SimL	SimM	AnL&M	SimH	AnH	SimL	SimM	AnL&M	SimH	AnH
(2, 2, 4)	0.8	0.554	0.525	0.483	0.545	0.545	1.204	1.070	1.034	1.018	1.009
	0.9	0.328	0.293	0.258	0.307	0.309	3.856	3.439	3.339	3.343	3.270
(4, 4, 6)	0.8	0.796	0.777	0.780	0.814	0.814	0.530	0.496	0.450	0.415	0.373
	0.9	0.546	0.506	0.500	0.546	0.547	2.451	2.355	2.230	2.175	2.039

Table 9
Expected inventory level and expected time to fulfill a backorder at a production facility.

(S, R_1, R_2)	ρ	$E[I^p]$					$E[W^{bp}]$				
		SimL	SimM	AnL&M	SimH	AnH	SimL	SimM	AnL&M	SimH	AnH
(2, 2, 4)	0.8	0.604	0.597	0.551	0.641	0.688	1.799	1.498	1.333	1.492	1.333
	0.9	0.320	0.311	0.274	0.336	0.370	3.819	3.244	3.000	3.214	3.000
(4, 4, 6)	0.8	1.792	1.782	1.840	2.051	2.151	1.737	1.482	1.333	1.488	1.333
	0.9	1.049	1.025	1.080	1.217	1.302	3.579	3.175	3.000	3.192	3.000

Table 10
Expected time spent by an order at a production facility.

(S, R_1, R_2)	ρ	$E[T^p]$				
		SimL	SimM	AnL&M	SimH	AnH
(2, 2, 4)	0.8	2.135	2.044	2.023	2.012	1.940
	0.9	3.900	3.626	3.559	3.560	3.405
(4, 4, 6)	0.8	3.021	2.999	3.000	2.944	2.915
	0.9	4.301	4.237	4.150	4.118	4.026

8 experiments for the case of Poisson arrivals and exponential processing times.

In order to evaluate the quality of our analytical results, we designed the following approach. For the fill rate (bounded by 1), we used the absolute difference, $|simulation - analytical|$, expressed as a percentage [23]. For the expected inventory level and the expected number of backorders, we used the normalized percentage error given by $100*(|simulation - analytical|)/base - stock\ level$ [12,19]. For the expected time to fulfill a backorder and the expected time spent by an order, we used the relative percentage error defined as $100*(|simulation - analytical|)/simulation$.

Tables 5–10 summarize the simulation results for LoIS (SimL), MedIS (SimM) and HiIS (SimH) cases along with the analytical results for HiIS (AnH) and the lower levels of inventory information sharing (AnL&M) at retail store 1 and a production facility. The results for the performance measures at retail store 2 are presented in an on-line supplemental material.

4.3. Performance of the analytical model for SCN under Poisson arrivals and exponential processing time distributions

The numerical results indicate that the analytical models developed for the different levels of inventory information sharing are fairly accurate. To be specific, the prediction error for the analytical model for SCN with HiIS is within 15% for a majority of the cases. Apart from being accurate for the SCN with HiIS, they also seem to yield reasonable bounds for the lower levels of inventory information sharing. Specifically, the model seems to yield upper bounds for fill rate and expected inventory level, and lower bounds for the expected number of backorders, expected time to fulfill a backorder, and expected time spent by an order at all stores. This can be attributed to the order routing strategies, which fully exploit the inventory information available from the production facility. For instance, when both production facilities have inventory, an order is routed to a production facility with equal probability in the case of LoIS and MedIS, while the orders are routed to the production facility with maximum inventory (or least number of outstanding orders) in the case of HiIS. When both facilities are backordered, the orders are routed based on fixed probability in the case of LoIS, while they are routed to the production facility with the least number of backorders in the case of MedIS and HiIS. Since the orders are always routed to the production facility with the least number of outstanding orders in the case of HiIS, we can see that this routing policy would result in the maximum value for fill rate and expected inventory level, and the lowest value for expected number of backorders, expected time to fulfill a backorder, and expected time spent by an order at all stores. Thus the analytical model for the HiIS case provides good approximation for the LoIS and MedIS cases. But the analytical model for the lower levels of inventory information sharing performs better in modeling the LoIS and MedIS cases for all performance measures at the retail stores and the production facilities (see Tables 5–10). We note that there is still an opportunity to develop additional correction factors to model the individual levels of information sharing using our knowledge about the routing policies. For example, it is known that the difference between these two routing policies, MedIS and LoIS, is observed when both the production facilities are backordered. In such instances, the orders are probabilistically routed in the LoIS case and based on the shortest queue in the MedIS case.

The analytical model can provide useful insights into the performance of the SCN with respect to production capacity and inventory related decisions. The following inferences were made based on the results from the numerical study:

1. As the level of information shared increases from LoIS to HiIS, there is an improvement in the performance measures at each store in the SCN. This can be seen from an increase in the fill rate and a decrease in the expected number of backorders. With a decrease in the expected number of backorders, we can expect to see an increase in the expected inventory level at all stores. The reason for this is the order routing strategy as explained earlier.
2. The improvement in the performance measures decreases with increasing base-stock levels as there is always enough inventory to fulfill customer demand. Also, the improvement decreases with increase in the utilization of the production facility and should vanish as the utilization tends to 1.

5. Analytical models for general inter-arrival and processing time distributions under different levels of inventory information sharing

In this section, we present the extensions of the analytical models for SCN with HiIS and the lower levels of inventory information sharing developed in Section 4. Section 5.1 presents the general model for the SCN under HiIS, while Section 5.2 presents the general model for lower levels of inventory information sharing.

5.1. General model for SCN with HiIS

We extend the pooled facility concept to the general case by modeling the two production facilities as a GI/G/2 queueing system. The expressions provided in Whitt [23] were used to obtain the distribution of the number of orders, N^{pp}, at the pooled

Table 11
Fill rate and expected number of backorders at retail store 1.

(ca_i^2, cs^2)	$f_1{}^r$					$E[B_1^r]$				
	SimL	SimM	AnL& M	SimH	AnH	SimL	SimM	AnL& M	SimH	AnH
$(S, R_1, R_2) = (4, 4, 6)$, $\rho = 80\%$										
(0.25, 0.25)	1.000	1.000	0.997	1.000	0.998	0.000	0.000	0.005	0.000	0.003
(0.25, 1.00)	0.987	0.987	0.977	0.991	0.985	0.025	0.023	0.053	0.016	0.035
(0.25, 4.00)	0.824	0.807	0.757	0.825	0.843	1.396	1.295	2.000	1.180	1.290
(1.00, 0.25)	0.981	0.982	0.980	0.986	0.986	0.031	0.029	0.041	0.022	0.030
(1.00, 1.00)	0.935	0.932	0.927	0.943	0.948	0.190	0.181	0.195	0.152	0.138
(1.00, 4.00)	0.771	0.755	0.716	0.770	0.808	2.018	1.835	2.523	1.723	1.707
(4.00, 0.25)	0.777	0.778	0.759	0.795	0.813	1.055	0.994	1.054	0.912	0.814
(4.00, 1.00)	0.723	0.719	0.691	0.734	0.765	1.683	1.617	1.716	1.529	1.301
(4.00, 4.00)	0.623	0.613	0.573	0.624	0.687	4.594	4.016	5.373	4.000	3.926
$(S, R_1, R_2) = (4, 4, 6)$, $\rho = 90\%$										
(0.25, 0.25)	0.990	0.989	0.978	0.994	0.983	0.017	0.016	0.059	0.009	0.044
(0.25, 1.00)	0.873	0.866	0.857	0.887	0.887	0.533	0.514	0.582	0.434	0.459
(0.25, 4.00)	0.577	0.544	0.478	0.565	0.586	6.797	6.242	7.508	6.009	5.952
(1.00, 0.25)	0.858	0.852	0.861	0.872	0.883	0.549	0.541	0.538	0.471	0.452
(1.00, 1.00)	0.730	0.721	0.706	0.743	0.755	1.757	1.691	1.763	1.567	1.468
(1.00, 4.00)	0.514	0.490	0.430	0.502	0.537	8.863	8.069	9.410	7.986	7.637
(4.00, 0.25)	0.491	0.486	0.440	0.504	0.508	6.027	5.931	6.405	5.699	5.635
(4.00, 1.00)	0.440	0.436	0.386	0.453	0.467	8.246	7.846	8.550	7.505	7.418
(4.00, 4.00)	0.369	0.355	0.304	0.367	0.414	17.147	15.174	17.782	15.180	14.954
$(S, R_1, R_2) = (4, 6, 4)$, $\rho = 80\%$										
(0.25, 0.25)	1.000	1.000	0.999	1.000	0.999	0.000	0.000	0.002	0.000	0.001
(0.25, 1.00)	0.996	0.995	0.989	0.997	0.993	0.008	0.008	0.026	0.005	0.017
(0.25, 4.00)	0.865	0.856	0.817	0.869	0.882	1.038	0.966	1.606	0.880	1.036
(1.00, 0.25)	0.993	0.993	0.991	0.995	0.994	0.011	0.011	0.019	0.008	0.014
(1.00, 1.00)	0.965	0.964	0.962	0.970	0.973	0.100	0.096	0.103	0.081	0.073
(1.00, 4.00)	0.816	0.809	0.781	0.821	0.852	1.587	1.425	2.055	1.339	1.390
(4.00, 0.25)	0.840	0.841	0.840	0.854	0.876	0.747	0.708	0.697	0.650	0.537
(4.00, 1.00)	0.786	0.787	0.778	0.798	0.831	1.269	1.229	1.231	1.162	0.932
(4.00, 4.00)	0.671	0.672	0.644	0.680	0.740	3.933	3.395	4.628	3.394	3.380
$(S, R_1, R_2) = (4, 6, 4)$, $\rho = 90\%$										
(0.25, 0.25)	0.997	0.997	0.988	0.998	0.991	0.005	0.005	0.031	0.003	0.023
(0.25, 1.00)	0.921	0.919	0.910	0.932	0.929	0.321	0.310	0.378	0.262	0.298
(0.25, 4.00)	0.624	0.605	0.552	0.623	0.645	5.903	5.399	6.576	5.203	5.214
(1.00, 0.25)	0.911	0.909	0.913	0.921	0.927	0.338	0.334	0.341	0.291	0.287
(1.00, 1.00)	0.798	0.795	0.784	0.810	0.820	1.293	1.246	1.295	1.157	1.078
(1.00, 4.00)	0.561	0.549	0.501	0.559	0.595	7.859	7.136	8.377	7.073	6.799
(4.00, 0.25)	0.559	0.559	0.526	0.574	0.583	5.168	5.102	5.417	4.900	4.766
(4.00, 1.00)	0.502	0.503	0.465	0.519	0.536	7.252	6.905	7.442	6.591	6.457
(4.00, 4.00)	0.407	0.403	0.360	0.414	0.462	15.896	14.014	16.475	14.042	13.855

production facility. The arrival process at the pooled output store is the superposition of the arrival streams from the two retail stores. The traffic rate equations and traffic variability equations (see [22]) shown in Eqs. (32) and (33) are used to obtain the mean and SCV (cap^2) of the arrival process at the pooled production facility. For details about the expression for the distribution of the number of orders at the pooled store (see Eq. (34)) and the expected number of orders at the pooled store ($E[N^{pp}]$), the reader is referred to Whitt [23]. The distribution of the number of orders at retail store i can be obtained using Eq. (41). The performance measures at retail store i can be obtained using Eqs. (4)–(9). The performance measures at a production facility can be obtained using Eqs. (35)–(40):

$$\lambda = \lambda_1^r + \lambda_2^r, \tag{32}$$

$$cap^2 = \gamma_1 \cdot ca_1^2 + \gamma_2 \cdot ca_2^2, \tag{33}$$

$$P(N^{pp} = k) = \begin{cases} P(C = k-2)P(Q^{pp} > 0); & k \geq 3 \\ p(k)(1 - P(Q^{pp} > 0)); & 0 \leq k \leq 2 \end{cases}, \tag{34}$$

where C is the random variable representing the conditional queue length given that the queue is non-empty and follows a geometric distribution as presented in Whitt [23], $P(Q^{pp} > 0)$ is the probability that the queue is non-empty, and

Table 12
Expected inventory level and expected time to fulfill a backorder at retail store 1.

(ca_i^2, cs^2)	$E[I_1^r]$					$E[W_1^{br}]$				
	SimL	SimM	AnL& M	SimH	AnH	SimL	SimM	AnL& M	SimH	AnH
$(S, R_1, R_2) = (4, 4, 6)$, $\rho = 80\%$										
(0.25, 0.25)	3.991	3.992	3.968	3.997	3.979	0.297	0.301	0.792	0.253	0.790
(0.25, 1.00)	3.804	3.817	3.821	3.875	3.881	0.955	0.862	1.171	0.862	1.169
(0.25, 4.00)	2.846	2.923	2.768	3.024	3.204	3.967	3.359	4.121	3.362	4.115
(1.00, 0.25)	3.801	3.819	3.835	3.864	3.880	0.845	0.790	1.056	0.783	1.054
(1.00, 1.00)	3.485	3.520	3.499	3.598	3.644	1.455	1.339	1.333	1.343	1.329
(1.00, 4.00)	2.631	2.737	2.589	2.821	3.043	4.403	3.743	4.445	3.754	4.437
(4.00, 0.25)	2.953	3.029	2.660	3.106	2.960	2.365	2.235	2.183	2.231	2.175
(4.00, 1.00)	2.697	2.782	2.393	2.854	2.777	3.042	2.875	2.772	2.879	2.762
(4.00, 4.00)	2.168	2.326	2.023	2.380	2.552	6.094	5.200	6.293	5.316	6.278
$(S, R_1, R_2) = (4, 4, 6)$, $\rho = 90\%$										
(0.25, 0.25)	3.817	3.824	3.838	3.898	3.880	0.805	0.749	1.313	0.740	1.313
(0.25, 1.00)	3.050	3.086	3.154	3.236	3.333	2.088	1.914	2.033	1.927	2.033
(0.25, 4.00)	1.762	1.872	1.646	1.969	2.134	8.026	6.846	7.194	6.899	7.194
(1.00, 0.25)	3.084	3.115	3.165	3.233	3.298	1.931	1.833	1.928	1.837	1.928
(1.00, 1.00)	2.484	2.551	2.497	2.667	2.749	3.257	3.029	3.000	3.047	3.000
(1.00, 4.00)	1.562	1.705	1.469	1.764	1.946	9.130	7.915	8.251	8.021	8.251
(4.00, 0.25)	1.809	1.901	1.455	1.976	1.761	5.914	5.778	5.723	5.757	5.723
(4.00, 1.00)	1.567	1.690	1.269	1.755	1.631	7.366	6.957	6.960	6.861	6.960
(4.00, 4.00)	1.170	1.334	1.022	1.386	1.496	13.586	11.753	12.758	11.988	12.758
$(S, R_1, R_2) = (4, 6, 4)$, $\rho = 80\%$										
(0.25, 0.25)	5.991	5.992	5.964	5.997	5.977	0.195	0.139	0.792	0.044	0.790
(0.25, 1.00)	5.789	5.802	5.795	5.865	5.864	0.916	0.854	1.210	0.852	1.208
(0.25, 4.00)	4.491	4.593	4.374	4.724	4.949	3.841	3.353	4.378	3.353	4.372
(1.00, 0.25)	5.786	5.800	5.813	5.851	5.864	0.820	0.787	1.088	0.776	1.086
(1.00, 1.00)	5.397	5.435	5.407	5.527	5.579	1.416	1.334	1.333	1.335	1.329
(1.00, 4.00)	4.175	4.327	4.121	4.437	4.726	4.311	3.727	4.693	3.742	4.686
(4.00, 0.25)	4.657	4.744	4.303	4.844	4.684	2.326	2.232	2.177	2.229	2.169
(4.00, 1.00)	4.279	4.394	3.908	4.487	4.408	2.972	2.881	2.767	2.880	2.758
(4.00, 4.00)	3.467	3.705	3.278	3.774	4.006	5.987	5.190	6.504	5.304	6.504
$(S, R_1, R_2) = (4, 6, 4)$, $\rho = 90\%$										
(0.25, 0.25)	5.805	5.812	5.811	5.891	5.859	0.780	0.738	1.325	0.721	1.325
(0.25, 1.00)	4.838	4.882	4.950	5.064	5.172	2.042	1.909	2.090	1.926	2.090
(0.25, 4.00)	2.868	3.029	2.715	3.164	3.395	7.856	6.834	7.340	6.888	7.340
(1.00, 0.25)	4.873	4.907	4.969	5.052	5.132	1.896	1.828	1.971	1.833	1.971
(1.00, 1.00)	4.020	4.107	4.029	4.257	4.359	3.198	3.030	3.000	3.047	3.000
(1.00, 4.00)	2.558	2.772	2.436	2.851	3.108	8.960	7.908	8.388	8.017	8.388
(4.00, 0.25)	2.949	3.073	2.467	3.177	2.892	5.859	5.791	5.720	5.762	5.720
(4.00, 1.00)	2.573	2.749	2.161	2.842	2.669	7.275	6.953	6.958	6.851	6.958
(4.00, 4.00)	1.919	2.175	1.715	2.248	2.396	13.423	11.741	12.877	11.992	12.877

$p(k) = \frac{e^{-\alpha}\alpha^k/k!}{\sum_{j=0}^{2} e^{-\alpha}\alpha^j/j!} = \frac{\alpha^k/k!}{\sum_{j=0}^{2} \alpha^j/j!}$ represents a truncated Poisson distribution (the Poisson density α can be obtained using the procedure described by Whitt [23, p. 156]:

$$f^p = \sum_{k=0}^{S^p-1} P(N^{pp} = k), \tag{35}$$

$$E[I^p] = 0.5 \sum_{k=0}^{S^p-1} (S^p - k)P(N^{pp} = k), \tag{36}$$

$$E[N^p] = 0.5 \sum_{k\geq 1} kP(N^{pp} = k), \tag{37}$$

$$E[B^p] = E[I^p] + E[N^p] - S, \tag{38}$$

$$E[W^{bp}] = \frac{E[B^p]}{(\lambda/2)(1 - f^p)}, \tag{39}$$

$$E[T^p] = \frac{S + E[B^p]}{\lambda/2}. \tag{40}$$

Table 13
Expected time spent by an order at retail store 1.

(ca_i^2, cs^2)	$E[T_1^r]$				
	SimL	SimM	AnL& M	SimH	AnH
$(S, R_1, R_2) = (4, 4, 6)$, $\rho = 80\%$					
(0.25, 0.25)	1.999	2.000	2.003	1.999	2.002
(0.25, 1.00)	2.012	2.011	2.026	2.007	2.018
(0.25, 4.00)	2.697	2.647	3.000	2.590	2.645
(1.00, 0.25)	2.017	2.015	2.021	2.012	2.015
(1.00, 1.00)	2.097	2.092	2.097	2.077	2.069
(1.00, 4.00)	3.011	2.919	3.261	2.863	2.854
(4.00, 0.25)	2.525	2.499	2.527	2.460	2.407
(4.00, 1.00)	2.844	2.812	2.858	2.768	2.650
(4.00, 4.00)	4.297	4.015	4.687	4.001	3.963
$(S, R_1, R_2) = (4, 4, 6)$, $\rho = 90\%$					
(0.25, 0.25)	2.008	2.008	2.029	2.004	2.022
(0.25, 1.00)	2.266	2.257	2.291	2.216	2.229
(0.25, 4.00)	5.397	5.120	5.754	5.003	4.976
(1.00, 0.25)	2.276	2.272	2.269	2.237	2.226
(1.00, 1.00)	2.881	2.847	2.881	2.785	2.734
(1.00, 4.00)	6.436	6.038	6.705	5.996	5.819
(4.00, 0.25)	5.015	4.969	5.202	4.856	4.818
(4.00, 1.00)	6.124	5.929	6.275	5.756	5.709
(4.00, 4.00)	10.586	9.589	10.891	9.597	9.477
$(S, R_1, R_2) = (4, 6, 4)$, $\rho = 80\%$					
(0.25, 0.25)	2.999	2.999	3.001	2.999	3.001
(0.25, 1.00)	3.003	3.003	3.013	3.002	3.009
(0.25, 4.00)	3.518	3.482	3.803	3.439	3.518
(1.00, 0.25)	3.008	3.007	3.010	3.006	3.007
(1.00, 1.00)	3.052	3.050	3.051	3.042	3.036
(1.00, 4.00)	3.796	3.715	4.028	3.672	3.695
(4.00, 0.25)	3.372	3.357	3.348	3.330	3.269
(4.00, 1.00)	3.637	3.619	3.616	3.586	3.466
(4.00, 4.00)	4.970	4.706	5.314	4.699	4.690
$(S, R_1, R_2) = (4, 6, 4)$, $\rho = 90\%$					
(0.25, 0.25)	3.002	3.002	3.015	3.001	3.012
(0.25, 1.00)	3.160	3.154	3.189	3.131	3.149
(0.25, 4.00)	5.950	5.698	6.288	5.601	5.607
(1.00, 0.25)	3.171	3.169	3.171	3.148	3.144
(1.00, 1.00)	3.649	3.626	3.648	3.580	3.539
(1.00, 4.00)	6.934	6.572	7.188	6.540	6.400
(4.00, 0.25)	5.585	5.556	5.708	5.458	5.383
(4.00, 1.00)	6.627	6.459	6.721	6.300	6.228
(4.00, 4.00)	10.961	10.010	11.238	10.029	9.927

$$P\left(N_i^r = m\right) = \sum_{l \geq m} P\left(N_i^r = m | B^{pp} = l\right) \cdot P\left(B^{pp} = l\right), \tag{41}$$

where $P\left(N_i^r = m | B^{pp} = l\right)$ and $P\left(B^{pp} = l\right)$ can be obtained using Eqs. (1)–(2).

5.2. General model for SCN with lower levels of inventory information sharing

The modification to obtain the general model for lower levels of inventory information sharing involves the refinement of the correction factor for the steady state probabilities (see Eq. (16)) developed for the SCN under Poisson arrivals and exponential processing times (see Section 4.2). The refinement takes into account the fact that the probability that there are no orders in the system depends on the variability of the inter-arrival and processing time distributions in addition to the base-stock level and the utilization of the servers. The scaled value of $P(N^{pp} = 0)$ is presented in Eq. (43):

$$\omega^G(\rho, S) = \frac{\exp^{(\rho^2 - \sqrt{\rho})c^2 S}}{2(1 - \rho^2)}, \tag{42}$$

where $c^2 = (cap^2 + cs^2)/2$ and cs^2 is the SCV of the processing time distribution.

Note that in the case of Poisson arrivals and exponential processing times, the term $\omega^G(\rho, S)$ in Eq. (42) reduces to the term $\omega^M(\rho, S)$ in Eq. (16):

$$P(N^{pp} = 0) = (1 - \rho)^2 \cdot \left(1 + \omega^G_{\rho,S}\right). \tag{43}$$

The distribution of the number of orders at the pooled production facility can be written as shown in Eq. (44). The normalization factor $\zeta^G(\rho, S)$ was then obtained as shown in Eq. (46). The probability $P(N^{pp} = 1)$ was fixed as $2\rho P(N^{pp} = 0)$, similar to the case of Poisson arrivals and exponential processing times. The rest of the probabilities were adjusted using the normalization factor as shown in the following equations:

$$P(N^{pp} = k) = \begin{cases} (1 - \rho)^2 \cdot \left(1 + \dfrac{\exp^{(\rho^2 - \sqrt{\rho})c^2 S}}{2(1 - \rho^2)}\right); & k = 0 \\ 2\rho P(N^{pp} = 0); & k = 1 \\ \dfrac{\alpha^2\left[1 - P\left(Q^{pp} > 0\right)\right]}{2\left(1 + \alpha + \alpha^2/2\right)} \zeta^G(\rho, S); & k = 2 \\ P\left(C = k - 2\right) P\left(Q^{pp} > 0\right) \zeta^G(\rho, S); & k \geq 3 \end{cases}, \tag{44}$$

$$P(N^{pp} = 0) = \left[1 + 2\rho\right] + \frac{\alpha^2\left[1 - P\left(Q^{pp} > 0\right)\right]}{2\left(1 + \alpha + \alpha^2/2\right)} \zeta^G(\rho, S) + P\left(Q^{pp} > 0\right) \zeta^G(\rho, S) \sum_{k \geq 1} P\left(C = k\right) = 1, \tag{45}$$

$$\zeta^G(\rho, S) = \frac{1 - P(N^{pp} = 0) \cdot \left(1 + 2\rho\right)}{\dfrac{\alpha^2 \cdot \left[1 - P\left(Q^{pp} > 0\right)\right]}{2 \cdot \left(1 + \alpha + \alpha^2/2\right)} + P\left(Q^{pp} > 0\right)}. \tag{46}$$

The distribution of the number of orders at retail store i can be obtained using Eq. (41). The performance measures at retail store i can be obtained using Eqs. (4)–(9), while those at a production facility can be obtained using Eqs. (35)–(40).

5.2.1. Validation of the analytical models

As before, the analytical results were compared to simulation estimates. The SCN parameter values used in the numerical experimentation are presented in Table 4. The design consists of 36 experiments.

Tables 11–16 summarize the simulation results for LoIS (SimL), MedIS (SimM) and HiIS (SimH) cases along with the analytical results for the HiIS case (AnH) and the lower levels of inventory information sharing (AnL&M) at retail store 1 and a production facility. The results for the performance measures at retail store 2 are presented in an on-line supplemental material.

5.3. Performance of the analytical model for SCN under general inter-arrival and processing time distributions

The numerical results indicate that the analytical models developed for the different levels of inventory information sharing are fairly accurate. To be specific, the prediction error for the

Table 14
Fill rate and expected number of backorders at a production facility.

(ca_i^2, cs^2)	f^p					$E[B^p]$				
	SimL	SimM	AnL& M	SimH	AnH	SimL	SimM	AnL& M	SimH	AnH
$(S, R_1, R_2) = (4, 4, 6)$, $\rho = 80\%$										
(0.25, 0.25)	0.991	0.991	0.976	0.997	0.984	0.007	0.006	0.028	0.002	0.019
(0.25, 1.00)	0.910	0.899	0.885	0.931	0.924	0.167	0.155	0.174	0.106	0.115
(0.25, 4.00)	0.721	0.678	0.553	0.708	0.711	1.923	1.783	2.424	1.619	1.565
(1.00, 0.25)	0.888	0.878	0.888	0.909	0.918	0.171	0.158	0.155	0.118	0.113
(1.00, 1.00)	0.796	0.777	0.739	0.814	0.814	0.530	0.496	0.522	0.415	0.370
(1.00, 4.00)	0.656	0.617	0.503	0.643	0.663	2.543	2.324	2.951	2.176	1.998
(4.00, 0.25)	0.596	0.572	0.466	0.606	0.585	1.575	1.473	1.795	1.354	1.390
(4.00, 1.00)	0.558	0.529	0.412	0.558	0.552	2.226	2.121	2.492	2.000	1.893
(4.00, 4.00)	0.513	0.478	0.376	0.498	0.542	4.791	4.253	5.513	4.200	4.031
$(S, R_1, R_2) = (4, 4, 6)$, $\rho = 90\%$										
(0.25, 0.25)	0.899	0.885	0.904	0.933	0.929	0.150	0.145	0.165	0.084	0.123
(0.25, 1.00)	0.693	0.655	0.627	0.712	0.705	1.113	1.073	1.071	0.900	0.845
(0.25, 4.00)	0.478	0.415	0.287	0.443	0.434	6.765	6.282	7.396	6.037	5.864
(1.00, 0.25)	0.662	0.627	0.626	0.677	0.686	1.103	1.070	1.029	0.929	0.866
(1.00, 1.00)	0.546	0.506	0.456	0.546	0.547	2.451	2.355	2.449	2.175	2.039
(1.00, 4.00)	0.418	0.366	0.253	0.384	0.393	8.532	7.778	8.955	7.668	7.290
(4.00, 0.25)	0.341	0.311	0.224	0.335	0.317	6.163	6.020	6.712	5.794	5.906
(4.00, 1.00)	0.317	0.287	0.194	0.307	0.301	8.045	7.613	8.461	7.310	7.340
(4.00, 4.00)	0.295	0.258	0.171	0.273	0.303	15.004	13.367	15.570	13.319	13.094

Table 15
Expected inventory level and expected time to fulfill a backorder at a production facility.

(ca_i^2, cs^2)	$E[I^p]$					$E[W^{bp}]$				
	SimL	SimM	AnL& M	SimH	AnH	SimL	SimM	AnL& M	SimH	AnH
$(S, R_1, R_2) = (4, 4, 6)$, $\rho = 80\%$										
(0.25, 0.25)	2.377	2.386	2.714	2.827	2.913	0.496	0.413	0.792	0.410	0.790
(0.25, 1.00)	1.992	1.989	2.117	2.383	2.528	1.227	1.029	1.008	1.028	1.006
(0.25, 4.00)	1.584	1.555	1.261	1.814	2.021	4.591	3.686	3.611	3.696	3.604
(1.00, 0.25)	2.066	2.072	2.062	2.368	2.422	1.020	0.860	0.919	0.860	0.917
(1.00, 1.00)	1.792	1.782	1.638	2.051	2.154	1.737	1.482	1.333	1.488	1.329
(1.00, 4.00)	1.489	1.470	1.121	1.660	1.870	4.935	4.042	3.961	4.066	3.952
(4.00, 0.25)	1.579	1.580	0.949	1.737	1.540	2.600	2.298	2.240	2.292	2.232
(4.00, 1.00)	1.463	1.451	0.860	1.596	1.508	3.357	3.007	2.824	3.017	2.815
(4.00, 4.00)	1.319	1.313	0.844	1.435	1.566	6.566	5.437	5.888	5.594	5.873
$(S, R_1, R_2) = (4, 4, 6)$, $\rho = 90\%$										
(0.25, 0.25)	1.676	1.670	2.022	2.173	2.326	0.986	0.837	1.153	0.835	1.153
(0.25, 1.00)	1.224	1.205	1.241	1.524	1.670	2.412	2.075	1.912	2.084	1.912
(0.25, 4.00)	0.897	0.862	0.572	1.029	1.148	8.644	7.152	6.912	7.219	6.912
(1.00, 0.25)	1.292	1.271	1.225	1.524	1.561	2.177	1.911	1.838	1.917	1.838
(1.00, 1.00)	1.049	1.025	0.888	1.217	1.302	3.597	3.175	3.000	3.192	3.000
(1.00, 4.00)	0.831	0.805	0.502	0.916	1.047	9.779	8.183	7.983	8.299	7.983
(4.00, 0.25)	0.859	0.840	0.425	0.938	0.804	6.238	5.825	5.767	5.812	5.767
(4.00, 1.00)	0.783	0.769	0.374	0.854	0.795	7.849	7.120	7.002	7.039	7.002
(4.00, 4.00)	0.709	0.687	0.344	0.762	0.846	14.200	12.014	12.524	12.213	12.524

analytical model for SCN with HiIS is within 10% for about 90% of the cases for all performance measures except the expected time to fulfill a backorder. This can be attributed to the fact that the expected time to fulfill a backorder is a conditional measure, and a small error in estimating the fill rate can result in a larger error in the measure. As in the case of the analytical model for Poisson arrivals and exponential processing times, the analytical model for HiIS serves as a good approximation for the lower levels of information sharing. But, the analytical models developed for the lower levels of information sharing using insights from simulation yield a better approximation for those levels.

The analytical models presented in this section can provide additional insights, especially on the effect of variability in the inter-arrival and processing times on the performance of the SCN:

Table 16
Expected time spent by an order at a production facility.

(ca_i^2, cs^2)	$E[T^p]$				
	SimL	SimM	AnL& M	SimH	AnH
$(S, R_1, R_2) = (4, 4, 6)$, $\rho = 80\%$					
(0.25, 0.25)	2.671	2.670	2.685	2.667	2.679
(0.25, 1.00)	2.777	2.770	2.782	2.736	2.744
(0.25, 4.00)	3.946	3.855	4.283	3.742	3.710
(1.00, 0.25)	2.782	2.772	2.770	2.746	2.742
(1.00, 1.00)	3.021	2.999	3.015	2.944	2.913
(1.00, 4.00)	4.362	4.218	4.634	4.118	3.999
(4.00, 0.25)	3.716	3.652	3.864	3.573	3.593
(4.00, 1.00)	4.153	4.085	4.328	4.002	3.928
(4.00, 4.00)	5.866	5.505	6.342	5.478	5.354
$(S, R_1, R_2) = (4, 4, 6)$, $\rho = 90\%$					
(0.25, 0.25)	2.766	2.763	2.777	2.722	2.749
(0.25, 1.00)	3.407	3.381	3.380	3.268	3.230
(0.25, 4.00)	7.175	6.853	7.597	6.691	6.576
(1.00, 0.25)	3.403	3.380	3.353	3.287	3.244
(1.00, 1.00)	4.301	4.237	4.300	4.118	4.026
(1.00, 4.00)	8.363	7.859	8.637	7.778	7.512
(4.00, 0.25)	6.780	6.682	7.141	6.535	6.604
(4.00, 1.00)	8.033	7.745	8.307	7.543	7.560
(4.00, 4.00)	12.679	11.585	13.047	11.550	11.396

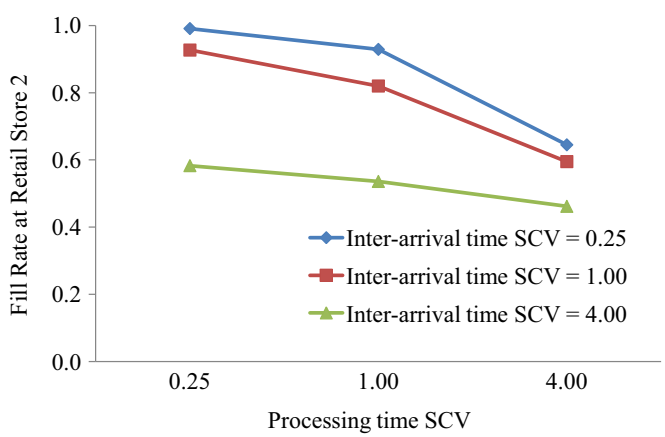

Fig. 4. Effect of processing time SCV.

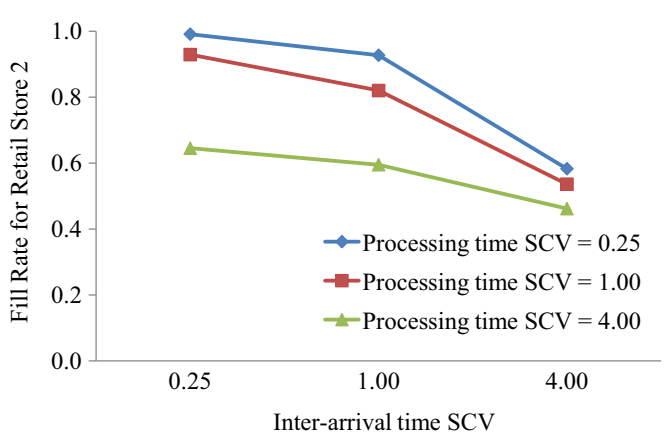

Fig. 5. Effect of inter-arrival time SCV.

Table 17
Experiments for the SCN configuration with transit delay.

Parameters	Levels	Parameter values
Demand arrival rates $(\lambda_1^r, \lambda_2^r)$	1	(2, 1)
Base-stock levels (S, R_1, R_2)	2	(3, 6, 3), (3, 3, 6)
Utilization level ρ	1	90%
Transit time distribution	2	Unif(1, 5), Unif(4, 8)
Inter-arrival time distribution at the retail stores	3	Erlang $(ca_i^2 = 0.25)$, Exponential $(ca_i^2 = 1.00)$, 2-phase hyper-exponential $(ca_i^2 = 4.00)$
Processing time distribution at the production facilities	3	Erlang $(cs^2 = 0.25)$, Exponential $(cs^2 = 1.00)$, 2-phase hyper-exponential $(cs^2 = 4.00)$

1. Increase in variability of the inter-arrival and processing times negatively impacts the SCN performance by decreasing the fill rate and increasing the expected number of backorders.
2. A more careful examination of the results presented in Table 11 reveals some interesting behavior. For instance, when either the inter-arrival times SCV or the processing time SCV is high (4 in our case), the fill rate at retail store 1 is less sensitive to changes in the other SCV (processing time SCV or inter-arrival time SCV). A similar behavior was also seen in the case of retail store 2 (results presented in on-line supplemental material). In Figs. 4 and 5 (results plotted for the HiIS case with a base-stock setting of (4, 6, 4) and a utilization of 90% at retail store 2), this decrease in sensitivity is illustrated by the bottom most plot becoming more horizontal. When the inter-arrival time SCV or the processing time SCV is fixed at a lower value (0.25 or 1 in our case), any change in the other SCV creates a more significant change in the fill rates at retail stores 1 and 2.

6. Analytical model for SCN with HiIS and transit delay under general inter-arrival and processing time distributions

We illustrate how the generalized analytical models developed in Section 5 can be enhanced to include transit delays. The modeling approach applies to SCNs with all levels of information sharing, but we demonstrate the performance prediction capability of the enhanced model only for the SCN with HiIS. We assume that the transit system behavior is independent of the demands and orders in the SCN and each transit system functions independent of another. The transit delays in the SCN from the production facilities to the retail stores are modeled using an infinite server queue as shown in Srivathsan [17].

The distribution of the number of orders at the pooled facility can be obtained using Eq. (34). The disaggregation procedure discussed in Section 4.1 was used to obtain the distribution of the number of orders from retail store i that are backordered at the pooled facility, Npp_i, as shown in the following equation:

$$P(Npp_i = m) = \sum_{l \geq m} P(N_i^r = m | B^{pp} = l) \cdot P(B^{pp} = l). \tag{47}$$

The conditional distribution $P(N_i^r = m | B^{pp} = l)$ is Binomial as shown in Eq. (1). The parameters of the binomial distributions are the proportions of orders from retail store i, γ_i, and l, the total number of backorders at the pooled facility.

When transit delay is considered, an order from a retail store can be either backordered or be in transit from the production facility to the retail store. Let N_i^r represent the random variable representing the number of orders at retail store i. The probability distribution of N_i^r was obtained in Eq. (48) by assuming that the random variables corresponding to the orders in transit, Tr_i, and

Table 18
Performance measures at retail store 1 (base-stock setting – (3, 6, 3)).

θ_i	(cs^2, ca_i^2)	f_1^r		$E[I_1^r]$		$E[B_1^r]$		$E[W_1^{br}]$		$E[T_1^r]$	
		Sim	Ana	Sim	Ana	Sim	Ana	Sim	Ana	Sim	Ana
3	(0.25, 0.25)	0.439	0.419	0.559	0.901	0.814	1.178	0.726	1.015	3.408	3.598
	(0.25, 1.00)	0.316	0.333	0.394	0.705	2.130	2.261	1.557	1.694	4.065	4.130
	(0.25, 4.00)	0.184	0.208	0.229	0.439	9.282	9.033	5.685	5.704	7.639	7.516
	(1.00, 0.25)	0.322	0.326	0.683	0.688	2.404	2.302	1.771	1.707	4.202	4.151
	(1.00, 1.00)	0.251	0.264	0.527	0.556	4.155	3.912	2.773	2.659	5.076	4.956
	(1.00, 4.00)	0.166	0.189	0.347	0.398	11.577	10.919	6.934	6.731	8.784	8.459
	(4.00, 0.25)	0.158	0.1597	0.803	0.332	9.288	9.140	5.517	5.439	7.643	7.570
	(4.00, 1.00)	0.139	0.150	0.706	0.313	11.965	11.052	6.940	6.502	8.976	8.526
	(4.00, 4.00)	0.107	0.146	0.544	0.307	19.341	18.708	10.824	10.947	12.665	12.354
6	(0.25, 0.25)	0.000	0.019	0.000	0.029	6.257	6.306	3.129	3.214	6.128	6.153
	(0.25, 1.00)	0.000	0.014	0.000	0.022	7.739	7.578	3.869	3.844	6.868	6.789
	(0.25, 4.00)	0.000	0.009	0.000	0.014	15.055	14.607	7.526	7.370	10.525	10.303
	(1.00, 0.25)	0.014	0.014	0.021	0.021	7.746	7.635	3.925	3.872	6.871	6.818
	(1.00, 1.00)	0.011	0.011	0.016	0.017	9.647	9.373	4.873	4.740	7.820	7.687
	(1.00, 4.00)	0.007	0.008	0.011	0.012	17.244	16.532	8.677	8.334	11.616	11.266
	(4.00, 0.25)	0.049	0.007	0.260	0.010	14.746	14.818	7.748	7.459	10.371	10.409
	(4.00, 1.00)	0.043	0.006	0.229	0.010	17.491	16.749	9.128	8.427	11.737	11.374
	(4.00, 4.00)	0.033	0.006	0.176	0.009	24.973	24.411	12.908	12.282	15.481	15.205

Table 19
Performance measures at retail store 1 (base-stock setting – (3, 3, 6)).

θ_i	(cs^2, ca_i^2)	f_1^r		$E[I_1^r]$		$E[B_1^r]$		$E[W_1^{br}]$		$E[T_1^r]$	
		Sim	Ana	Sim	Ana	Sim	Ana	Sim	Ana	Sim	Ana
3	(0.25, 0.25)	0.014	0.057	0.006	0.057	3.263	3.353	1.655	1.778	3.871	3.806
	(0.25, 1.00)	0.010	0.043	0.003	0.035	4.742	4.613	2.394	2.411	7.527	7.314
	(0.25, 4.00)	0.005	0.027	0.056	0.055	12.057	11.629	6.060	5.974	3.888	3.835
	(1.00, 0.25)	0.042	0.042	0.042	0.044	4.777	4.669	2.493	2.437	4.833	4.700
	(1.00, 1.00)	0.032	0.036	0.027	0.032	6.670	6.401	3.442	3.312	8.625	8.276
	(1.00, 4.00)	0.021	0.024	0.228	0.025	14.257	13.552	7.276	6.944	7.355	7.417
	(4.00, 0.25)	0.049	0.020	0.198	0.024	11.713	11.834	6.154	6.035	8.722	8.382
	(4.00, 1.00)	0.042	0.019	0.150	0.025	14.457	13.764	7.540	7.012	12.468	12.213
	(4.00, 4.00)	0.032	0.019	0.000	0.001	21.947	21.426	11.33	10.917	6.128	6.139
6	(0.25, 0.25)	0.000	0.000	0.000	0.000	9.257	9.278	4.627	4.641	6.128	6.139
	(0.25, 1.00)	0.000	0.000	0.000	0.000	10.738	10.556	5.368	5.280	6.868	6.778
	(0.25, 4.00)	0.000	0.000	0.000	0.000	18.055	17.594	9.025	8.799	10.525	10.297
	(1.00, 0.25)	0.000	0.000	0.000	0.000	10.725	10.615	5.361	5.209	6.860	6.807
	(1.00, 1.00)	0.000	0.000	0.000	0.000	12.631	12.357	6.314	6.180	7.812	7.678
	(1.00, 4.00)	0.000	0.000	0.000	0.000	20.233	19.520	10.113	9.762	11.611	11.260
	(4.00, 0.25)	0.014	0.000	0.065	0.000	17.551	17.808	8.895	8.906	10.273	10.404
	(4.00, 1.00)	0.012	0.000	0.057	0.000	20.319	19.740	10.273	9.870	11.651	11.370
	(4.00, 4.00)	0.009	0.000	0.043	0.000	27.840	27.402	14.041	13.703	15.414	15.201

those backordered at the production facility, Npp_i, are independent.

$$P(N_i^r = n) = \sum_{l=0}^{n} P(Npp_i = l)P(Tr_i = n - l) \quad n = 0, 1, 2, \ldots. \tag{48}$$

As the transit delay is modeled as an M/G/∞ queueing system, the random variable Tr_i is Poisson distributed as shown in Eq. (49). Note that the total arrival rate at the transit node is equal to the external arrival rate at the individual retail stores (λ_i^r) and θ_i is the mean transit time from the production facilities to retail store i:

$$P(Tr_i = x) = \frac{e^{-\delta_i}\delta_i^x}{x!} \quad x = 0, 1, 2, \ldots, \tag{49}$$

where

$$\delta_i = \lambda_i^r \theta_i. \tag{50}$$

The performance measures at retail store i can be obtained using Eqs. (4)–(9), while those at a production facility can be obtained using Eqs. (35)–(40). The accuracy of the analytical model was evaluated by comparing the analytical results (Ana) to simulation estimates (Sim). The SCN parameter values used in the numerical experimentation are presented in Table 17. The design consists of 36 experiments.

The results at retail store 1 and a production facility are presented in Tables 18–20. The results for the performance measures at retail store 2 are presented in the on-line supplemental material. The summary of results is presented in Table 21.

More than 90% of all performance measures except average number of backorders fall within a 15% error range (see Table 21). The analytical model follows the trend in the different performance measures as indicated by the simulation results. In addition, it was also observed that an increase in transit times resulted

Table 20
Performance measures at a production facility (base-stock setting – (3, 6, 3)).

θ_i	(cs^2, ca_i^2)	f^p		$E[I^p]$		$E[B^p]$		$E[W^{bp}]$		$E[T^p]$	
		Sim	Ana	Sim	Ana	Sim	Ana	Sim	Ana	Sim	Ana
3	(0.25, 0.25)	0.845	0.855	1.283	1.412	0.192	0.208	0.826	0.960	2.127	2.139
	(0.25, 1.00)	0.591	0.584	0.878	0.992	1.304	1.167	2.123	1.868	2.869	3.175
	(0.25, 4.00)	0.379	0.370	0.638	0.729	6.795	6.445	7.285	6.818	6.525	6.297
	(1.00, 0.25)	0.551	0.553	0.891	0.905	1.290	1.211	1.916	1.805	2.860	2.807
	(1.00, 1.00)	0.436	0.441	0.712	0.781	2.718	2.517	3.208	3.000	3.809	3.678
	(1.00, 4.00)	0.325	0.334	0.576	0.668	8.413	7.890	8.297	7.895	7.596	7.260
	(4.00, 0.25)	0.258	0.246	0.577	0.504	6.354	6.606	5.707	5.840	6.236	6.404
	(4.00, 1.00)	0.240	0.241	0.529	0.509	8.421	8.055	7.388	7.073	7.619	7.370
	(4.00, 4.00)	0.221	0.261	0.481	0.554	14.066	13.801	12.002	12.446	11.349	11.201
6	(0.25, 0.25)	0.845	0.855	1.283	1.412	0.192	0.208	0.826	0.960	2.127	2.139
	(0.25, 1.00)	0.591	0.584	0.878	0.992	1.304	1.167	2.123	1.868	2.869	3.175
	(0.25, 4.00)	0.379	0.370	0.638	0.729	6.795	6.445	7.285	6.818	6.525	6.297
	(1.00, 0.25)	0.551	0.553	0.891	0.905	1.290	1.211	1.916	1.805	2.860	2.807
	(1.00, 1.00)	0.436	0.441	0.712	0.781	2.718	2.517	3.208	3.000	3.809	3.678
	(1.00, 4.00)	0.325	0.334	0.576	0.668	8.413	7.890	8.297	7.895	7.596	7.260
	(4.00, 0.25)	0.258	0.246	0.577	0.504	6.354	6.606	5.707	5.840	6.236	6.404
	(4.00, 1.00)	0.240	0.241	0.529	0.509	8.421	8.055	7.388	7.073	7.619	7.370
	(4.00, 4.00)	0.221	0.261	0.481	0.554	14.066	13.801	12.002	12.446	11.349	11.201

Table 21
Summary of results for SCN with transit delay.

Error range (%)	Percentage of results within error range				
	Fill rate	Expected number of backorders	Expected inventory level	Expected time to fulfill a backorder	Expected time spent by an order
<5	94.4	45.4	79.6	62.0	89.8
<10	100.0	70.4	98.1	84.3	96.3
<15	100.0	88.0	100.0	91.7	100.0
<20	100.0	97.2	100.0	95.4	100.0
<25	100.0	100.0	100.0	95.4	100.0

in a decrease in fill rates and expected inventory levels and an increase in the other three performance measures at all stages.

7. Conclusions and ongoing work

We developed analytical models for three different levels of upstream inventory information sharing, and they were found to accurately follow the trends in the different performance measures in our numerical study.

The analytical models are most useful in assessing the marginal improvements in the SCN performance as additional inventory-related information is shared:

1. First, as expected, increase in the utilization level at the production facilities, increase in the variability of the inter-arrival and processing times, increase in the mean transit times, or a decrease in the base-stock levels negatively impacted the SCN performance.
2. Second, the analytical models were also useful in gaining some additional key insights into the SCN performance. It was observed that when either the inter-arrival time SCV or the processing time SCV was high, the fill rates at the two retail stores were less sensitive to changes in the other SCV.
3. Finally, it was also demonstrated that sharing more information (LoIS to HiIS) results in an increase in fill rates and expected inventory levels along with a decrease in the expected number of backorders, expected time to fulfill a backorder, and the expected time spent by an order. This can be attributed to the order routing strategies used by the retail store.

Our ongoing work focuses on generalizing the models presented in this paper and modeling more general SCN configurations as outlined below.

1. *Modeling asymmetry in the production facilities:* We have successfully modeled asymmetry at the retail store level by considering different base-stock levels and different demand arrival rates in this study. However, there is a need to consider asymmetry in the production facilities. Asymmetry in the production facilities could be due to various aspects such as different base-stock levels, different processing time distributions, difference in ordering preferences, or a combination of the above factors.
   - *Different base-stock levels:* When this situation occurs in our SCN with HiIS, the initial orders would be routed to the production facility with the higher base-stock level and the order routing would resemble that in a shortest queue only when the inventory level at the production facility with higher base-stock level becomes equal to the base-stock level of the other facility. This observation could be used to develop an approximation for the distribution of the number of orders at the pooled output store.
   - *Different processing time distributions:* Another source of asymmetry in the production facilities is different processing time distributions. In this case, a possible procedure would be to replace the two asymmetric servers by two identical servers with an aggregate processing time distribution, and the pooled facility could be modeled as a multi-server queueing system with general processing time distribution.
   - *Difference in preference levels:* In SCNs with virtual pooling, a higher priority may have to be given to a production facility that is closer to the retail store that placed the order. In such cases, the probability of a retail store placing an order at a production facility would not be identical as is the case in our current research. This could result in different utilizations at the production facilities even when they have identical base-stock levels and processing time distributions. In such cases, the marginal distribution of the number of orders at each facility can be obtained by conditioning on the number of orders at the pooled facility.

2. *Modeling more complex SCNs with inventory information sharing:* The study presented in this paper considered analytical models for "building-block" type configurations. We believe that there is a need to extend these models for more general SCN structures. This is also a focus of our ongoing work:
   - *Modeling additional retail stores:* In this case, the modeling approach would be very similar to that used in this paper. However, the conditional distributions used in the "disaggregation" procedure would follow a multinomial distribution instead of a binomial distribution.
   - *Modeling additional production facilities:* The pooled production facility can be modeled as a GI/G/m queueing system, where m represents the number of production facilities.
   - *Modeling additional echelons:* In order to model additional echelons, a methodology using the parametric decomposition approach similar to the work contained in Srivathsan and Kamath [19] could be explored.

Acknowledgment

We would like to thank the anonymous reviewers for their comments and suggestions that have helped us in improving our paper. We would also like to thank Ms. Pam Reynolds for assisting us by reviewing the revised manuscript, and making several suggestions that improved the writing.

Appendix A. Supplementary data

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.cor.2016.07.011.

References

[1] Adan I, Wessels J, Zijm W, Nederlandse Philips Bedrijven B. Analysis of the symmetric shortest queue problem. Stoch Models 1990;6:691–713.
[2] Armony M, Plambeck E. The impact of duplicate orders on demand estimation and capacity investment. Manag Sci 2005;51(10):1505–18.
[3] Buzacott J, Shanthikumar J. Stochastic models of manufacturing systems. Englewood Cliffs, NJ, USA: Prentice Hall, Inc.; 1993.
[4] Corsten D, Gruen T. Desperately seeking shelf availability: an examination of the extent, the causes, and the efforts to address retail out-of-stocks. Int J Retail & Distrib Manag 2003;31(12):605–17.
[5] Ettl M, Feigin G, Lin G, Yao D. A supply network model with base-stock control and service requirements. Oper Res 2000;48(2):216–32.
[6] Gavirneni S. Benefits of cooperation in a production distribution environment. Eur J Oper Res 2001;130(3):612–22.
[7] Gross D, Shortle JF, Thompson JM, Harris C. Fundamentals of queueing theory. 4th edition. Hoboken, New Jersey: John Wiley & Sons, Inc.; 2008.
[8] Halfin S. The shortest queue problem. J Appl Probab 1985;22(4):865–78.
[9] Kamath M, Srivathsan S. A comparative evaluation of analytical approximations for production-inventory networks. Comput Ind Eng 2012;62:644–52.
[10] Kelton W, Sadowski R, Sadowski D. Simulation with arena. 2nd edition. New York, NY, USA: The McGraw-Hill Companies, Inc.; 2002.
[11] Kingman J. Two similar queues in parallel. Ann Math Stat 1961;32:1314–23.
[12] Krishnamurthy A, Suri R. Performance analysis of single stage Kanban controlled production systems using parametric decomposition. Queueing Syst 2006;54:141–62.
[13] Lee H, Billington C. Material management in decentralized supply chains. Oper Res 1993;41(5):835–47.
[14] Lee Y-J, Zipkin P. Tandem queues with planned inventories. Oper Res 1992;40(5):936–47.
[15] Li G, Yan H, Wang S, Xia Y. Comparative analysis on value of information sharing in supply chains. Supply Chain Manag 2005;10(1):34–46.
[16] Sivaramakrishnan S. Analytical models for performance evaluation of production-inventory systems [PhD dissertation]. Stillwater, OK, USA: Oklahoma State University; 1998.
[17] Srivathsan S. Analytical performance modeling of supply chain networks [Master's thesis]. Stillwater, OK, USA: Oklahoma State University; December 2004.
[18] Srivathsan S. Modeling inventory information visibility in supply chain networks [PhD thesis]. Stillwater, OK: Oklahoma State University; 2012.
[19] Srivathsan S, Kamath M. An analytical performance modeling framework for supply chain networks. IEEE Trans Autom Sci Eng 2012;9(2):265–75.
[20] Srivathsan S, Kamath M. Performance modeling of a two-echelon supply chain with inventory visibility. In: Proceedings of the 2012 industrial and systems engineering research conference (ISERC 2012), Orlando, FL, USA; 2012. p. 1–10.
[21] Welch P. The statistical analysis of simulation results. In: Lavenberg SS, editor. The computer performance modeling handbook. New York, NY, USA: Academic Press; 1983. p. 268–328.
[22] Whitt W. The queueing network analyzer. Bell Syst Tech J 1983;62(9):2779–815.
[23] Whitt W. Approximations for the GI/G/m queue. Prod Oper Manag 1993;2(2):114–61.
[24] Wolff R. Poisson arrivals see time averages. Oper Res 1982;30(2):223–31.
[25] Zipkin P. Performance analysis of a multi-item production-inventory system under alternative policies. Manag Sci 1995;41(4):690–703.